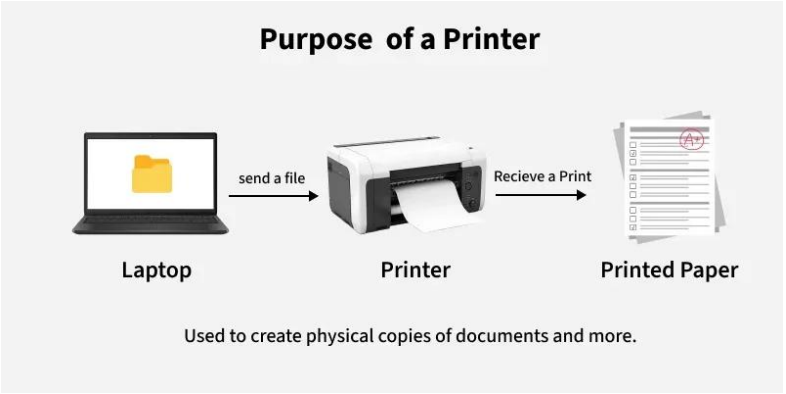
Understanding Printers

Working Principles, Printing Technologies, Maintenance, Troubleshooting, and Secure Printing Practices



Lovnish Verma

Learning Outcomes



- Explain what a printer is and classify common printer types
- Identify major components and trace a print job end to end
- Describe how inkjet, laser, dot-matrix and thermal printers form an image
- Interpret specifications: PPM, IPM, DPI, duty cycle, yield
- Apply a structured 8-step method to common faults
- Apply secure printing practices and recognise network printer risks

By the end of the session

Participants can choose a suitable printer technology, perform user-level checks safely, and recognise when to escalate to a technician.

What Is a Printer?



- An output device that converts digital data into a physical representation on paper or other media
- Hard copy: tangible, persistent output (paper, labels, forms)
- Soft copy: on-screen or electronic output (display, PDF)
- Applications: reports, forms, drawings, labels, receipts, asset tags
- Government use: records, correspondence, controlled documents

Hard copy vs soft copy

Hard copy persists without power and can be handled, signed or filed.
Soft copy is easily edited and distributed but depends on a device. Printed copies of sensitive material also need physical handling controls.

Classification of Printers



Impact

Form image by striking a ribbon against paper. Example: dot-matrix, line printers.

Non-impact

Form image without striking. Examples: inkjet, laser (electrophotographic), thermal.

Standalone (single-function)

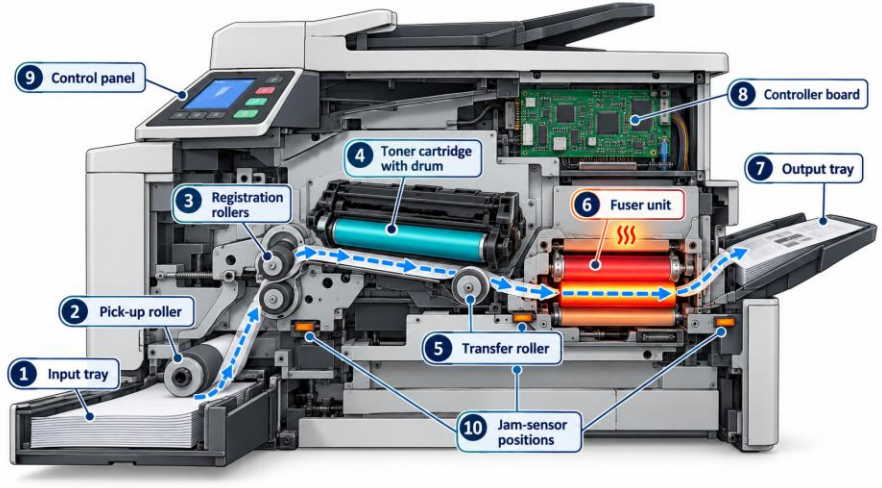
Print only. Connected by USB or network.

Multifunction (MFP)

Print, copy, scan and often fax from one device on the network.

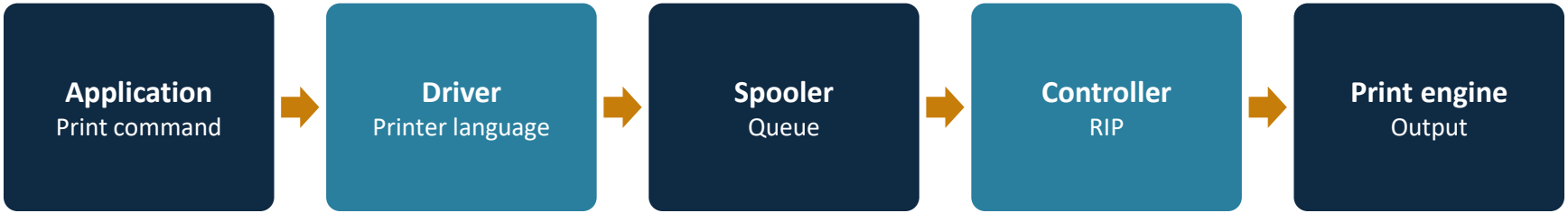
Anatomy of a Printer

- Paper trays and feed path: store and move media
- Print engine: forms the image (drum, fuser or print head)
- Controller board and firmware: process data, run the engine
- Control panel: status, settings, authentication
- Sensors: paper presence, jams, door, toner/ink level



The Printing Workflow

- Application sends document to the operating system print subsystem
- Driver converts it to a language the printer understands
- Spooler queues the job, so applications are not blocked
- Controller rasterises the page (raster image processing) and drives the engine



Drivers, Spoolers, and Print Servers



- **Driver:** software translating application output for a specific printer model
- **Spooler:** stores jobs on disk and feeds them in order
- **Print server:** shared host managing queues for many clients
- **Local queue:** job spooled on the user's own computer
- **Network (server) queue:** job held on a central server; central control of drivers, access and logs

Why central queues help

Centralised queues allow consistent drivers, access control, auditing and secure-release integration. Stuck jobs in a queue are a frequent cause of 'printer not responding'.

Overview of Printing Technologies

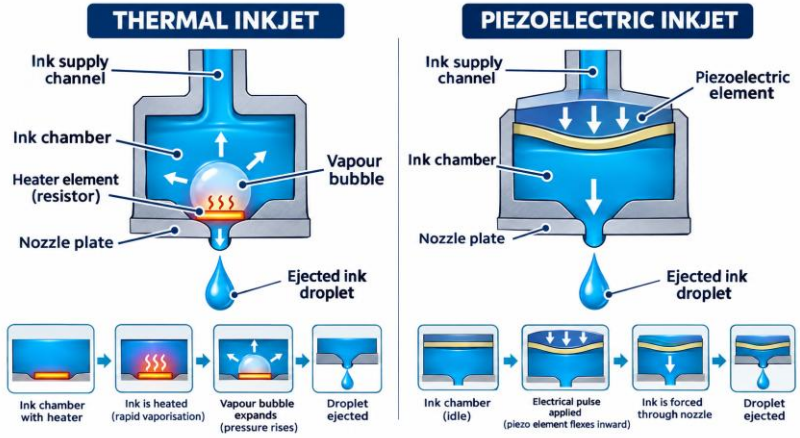


Technology	Mechanism	Speed	Running cost	Noise
Inkjet	Ejects liquid ink droplets	Low to medium	Medium to high per page	Low
Laser	Toner fused by heat (electrophotography)	Medium to high	Low per page	Low to medium
Dot-matrix	Pins strike an inked ribbon	Low to medium	Low	High
Thermal	Heat on coated paper or ribbon	Medium to high	Low (labels, receipts)	Low

Ratings are general, relative comparisons; actual values depend on model.

Inkjet Printers

- Print head fires microscopic ink droplets through nozzles onto paper
- Thermal inkjet: a heating element vaporises ink, forming a bubble that expels a droplet
- Piezoelectric inkjet: an electric pulse deforms a piezo element to push out a droplet
- Suited to low-volume colour and photo output
- Nozzles can clog if unused; head cleaning cycles consume ink



Laser Printers: The 7 Stages

- Laser printers use electrophotography
- A photosensitive drum is charged, written on by light, dusted with toner, then the toner is transferred and fused

1 Charging
Uniform charge on drum

2 Exposing
Light writes latent image

3 Developing
Toner clings to image

4 Transferring
Toner moves to paper

5 Fusing
Heat and pressure bond it

6 Cleaning
Residual toner removed

7 Erasing
Drum charge neutralised

Inside a Laser Printer



- **Drum:** photosensitive surface that holds the latent image
- **Toner:** fine powder of pigment and polymer, charged to be attracted to the image
- **Fuser:** heated roller and pressure roller that melt toner into paper fibres
- **Relationship:** drum creates the pattern, toner forms the image, fuser makes it permanent

Analogy: static cling

Rub a balloon on cloth and it attracts paper scraps. The drum behaves similarly: charged regions attract toner powder, uncharged regions do not. The fuser then acts like an iron, melting the powder in place.

Dot-Matrix Printers



- Print head holds a column of pins, commonly 9 or 24
- Electromagnets fire pins against an inked ribbon and paper
- Impact allows multipart carbon or carbonless forms
- Low cost per page, but noisy and low resolution
- Still used for continuous stationery and multipart records

Key point

Because the head physically strikes the paper, dot-matrix is the one common technology here that can print through several copies at once.

Thermal Printers



- Direct thermal: heat darkens special heat-sensitive paper; no ink or toner
- Thermal transfer: heat melts wax or resin from a ribbon onto the label
- Direct thermal output can fade with heat, light and age
- Thermal transfer output is more durable
- Use cases: asset labels, barcodes, receipts, shipping labels

Choosing between them

Short-life items such as receipts suit direct thermal. Asset tags and labels that must last suit thermal transfer.

Multifunction Printers (MFPs)



- Combine print, copy, scan and optional fax
- Automatic document feeder (ADF) handles multi-page originals
- Flatbed glass handles single, bound or fragile originals
- Network interface enables scan-to-email and shared printing
- Internal storage and services widen the security footprint

MFP = printer + scanner + network host

Because an MFP runs firmware, stores data and connects to the network, it should be managed as an IT asset, not just office equipment.

Technology Comparison Matrix



Criterion	Inkjet	Laser	Dot-matrix	Thermal
Cost per page	Medium to high	Low	Low	Low
Print quality	High for photos	Sharp text	Basic	Good for barcodes
Speed	Low to medium	Medium to high	Low to medium	Medium to high
Typical workload	Home, low-volume colour	Office documents	Multipart forms	Labels, receipts
Consumable	Ink cartridges	Toner, drum, fuser	Ribbon	Paper or ribbon

Printer Specifications



Term	Meaning
PPM	Pages per minute; rated print speed
IPM	Images per minute; counts sides, relevant to duplex
DPI	Dots per inch; addressable resolution
Duty cycle	Maximum recommended monthly page volume
Yield	Pages a consumable is rated to print

Why DPI isn't everything

Output quality also depends on toner or ink quality, paper, halftoning and the image processing in the controller. Higher DPI alone does not guarantee better text.

Connectivity and Protocols



- USB: direct, single computer
- Ethernet: wired network; preferred for stability
- Wi-Fi: convenient but needs strong wireless security
- IPP: Internet Printing Protocol; IPPS adds TLS
- Also common: LPR/LPD and raw port 9100 printing, usually unencrypted
- Disable unused protocols and services

Secure network printing basics

Prefer encrypted protocols (IPPS), place printers on a controlled network segment, restrict who can reach the printer, and turn off unneeded services.

Common Symptoms



Paper jams

Misfeed, wrong media, worn rollers

Faded output

Low toner or ink, density setting

Streaks or lines

Dirty or damaged drum, clogged nozzle

Smudges

Fuser or ink-drying issue, wrong media

Network failure

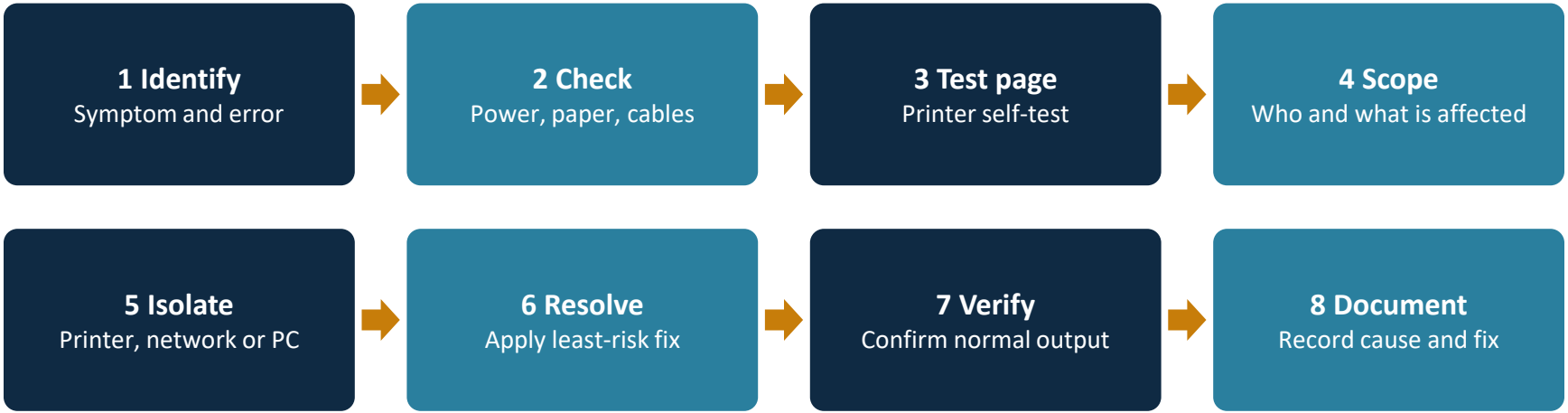
Offline, IP change, stuck queue

No output

Queue paused, driver, power, cable

The 8-Step Troubleshooting Method

- Work from simple to complex; change one thing at a time
- Decision point at Scope: one user, many users, or one document?



Jams and Quality Defects



- Switch off and disconnect power before reaching into the paper path
- Let the fuser cool; it remains hot after printing
- Remove jammed paper gently in the paper-travel direction; avoid tearing
- Never touch high-voltage parts or force sealed covers
- Streaks, ghosting or repeated marks: record the pattern, then escalate

User-level vs technician-level

User-level: clear accessible jams, replace consumables, clean exterior and glass per the manual. Technician-level: replace fuser or drum assemblies, open sealed or high-voltage modules, board repair.

Preventive Maintenance



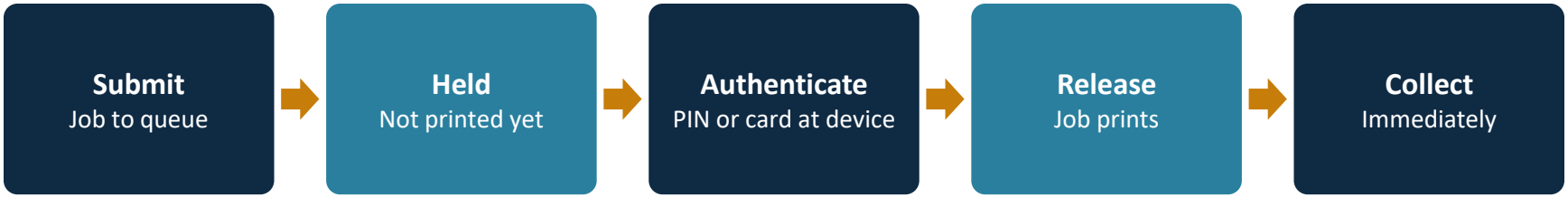
- Clean per manufacturer instructions: exterior, scanner glass, ADF rollers
- Apply firmware updates through the approved process
- Store paper flat, sealed and dry to avoid curl and jams
- Use compatible, approved consumables
- Dispose of cartridges and toner through approved recycling or waste channels
- Keep a maintenance log

Toner handling

Do not blow toner dust into the air or use an ordinary household vacuum on spills; follow the manufacturer's guidance and use a toner-rated vacuum or damp cloth.

Secure Printing

- Secure release: job is held until the user authenticates at the device (PIN or card)
- Retrieve printouts immediately; never leave output on the tray
- Handle printed material according to its classification
- Use the correct printer for the classification level
- Destroy unneeded copies using approved methods



Network Printer Risks and Mitigations



Risk	Practical mitigation
Default credentials	Change admin passwords at commissioning; limit admin access
Unencrypted traffic	Use IPPS/TLS; restrict protocols and ports
Retained jobs on MFP storage	Enable encryption and secure erase; set job retention limits
Unpatched firmware	Patch through approved change control
Open network exposure	Segment printers; allow only required hosts
Unattended output	Secure release; collect immediately

Live Demonstration Plan



- Open the document and choose Print (Ctrl+P on Windows or Linux, Cmd+P on macOS)
- Select the correct printer from the list
- Open preferences: set copies, orientation, paper size, duplex, colour or mono
- Check the print preview
- Print, then confirm output at the printer
- If it fails: open the queue and check status

Troubleshooting Scenarios



A: Stuck queue

Symptom: jobs sit in the queue; nothing prints.
Check: printer online, queue paused or errored.
Action: cancel the stuck job, restart the spooler service, resend.

B: Vertical marks

Symptom: vertical lines on every page.
Check: print a self-test page; inspect scanner glass if copying.
Action: clean accessible parts; if it persists, escalate.

C: Network drop

Symptom: printer shows offline.
Check: cable or Wi-Fi, IP address, ping, other users.
Action: restore link; correct the port or IP; escalate if network-wide.

Knowledge Check (Q1 to Q4)



Q1. Which laser printer component bonds toner to paper?

A. Drum
B. Fuser
C. Developer roller
D. Spooler

Q2. What is the correct order of the laser printing stages?

A. Charge, Expose, Develop, Transfer, Fuse, Clean, Erase
B. Expose, Charge, Develop, Fuse, Transfer, Clean, Erase
C. Charge, Develop, Expose, Transfer, Fuse, Erase, Clean
D. Develop, Charge, Expose, Fuse, Transfer, Erase, Clean

Q3. Which software component queues jobs until the printer is ready?

A. Driver
B. RIP
C. Firmware
D. Spooler

Q4. Which technology can print through multipart carbon forms?

A. Inkjet
B. Direct thermal
C. Dot-matrix
D. Laser

Knowledge Check (Q5 to Q8)



Q5. A printer is rated 1200 DPI. This describes its:

A. Pages per minute
B. Print resolution
C. Duty cycle
D. Cartridge yield

Q6. Vertical streaks on every laser page most likely indicate:

A. Worn or dirty imaging parts or paper path
B. Spooler stopped
C. Wrong IP address
D. Low paper

Q7. Which practice best reduces unattended sensitive output?

A. Larger paper tray
B. Duplex by default
C. Secure release printing
D. Wi-Fi only

Q8. Default administrator credentials on a network printer should be:

A. Left unchanged
B. Shared by the team
C. Printed on a cover sheet
D. Changed at commissioning

Key Takeaways



- A printer converts digital data to hard copy via driver, spooler, controller and engine
- Laser printing follows seven electrophotographic stages
- Technology choice depends on workload, cost and media
- Read specifications critically; DPI alone is not quality
- Troubleshoot methodically; clear only user-level faults, escalate the rest
- Secure printing and hardened network printers protect information

Remember

Safety first around hot and high-voltage parts. Follow your organisation's printing and classification policy.

Questions and Discussion



Q&A

Questions, scenarios and local policy clarifications

Appendix A: Answer Key

Q	Answer	Rationale
1	B	Fuser applies heat and pressure
2	A	Standard seven-stage sequence
3	D	Spooler stores and orders jobs
4	C	Impact printing strikes through copies
5	B	DPI is print resolution
6	A	Imaging parts or path contamination
7	C	Release only after authentication
8	D	Defaults are widely known

Appendix B: References



- ISO/IEC 19752: yield of monochrome electrophotographic toner cartridges
- ISO/IEC 19798 and ISO/IEC 24711: colour laser and inkjet cartridge yield
- RFC 8010 and RFC 8011: Internet Printing Protocol/1.1
- Printer Working Group (PWG) printing standards
- NIST SP 800-53: general security and privacy controls
- Manufacturer user, safety and hardening guides for devices in use

Thank You….